BLAST RESEARCH POSTER — ABSTRACT

Keys Without Trust Assumptions: Integrating Quantum Key Distribution and Post-Quantum Cryptography into the Internet Protocol Stack

A CN1 analysis of the QKD control/data plane, the ETSI key-delivery API, and hybrid IKEv2 key establishment for quantum-era networks

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ABSTRACT

Modern network security relies on key-establishment mechanisms whose security depends on mathematical assumptions that a sufficiently capable quantum computer could undermine. This creates a "harvest-now, decrypt-later" concern for information requiring long-term confidentiality. This project examines Quantum Key Distribution (QKD) integrated with Post-Quantum Cryptography (PQC), framing quantum-era security as a computer-networking integration challenge rather than solely a physics problem.

QKD uses quantum-channel measurement statistics to generate shared symmetric key material; interception can affect the quantum bit error rate and trigger rejection of an unsafe key. The selected design treats QKD as a key source alongside PQC, not as an encryption algorithm or replacement for IP. It maps a three-plane architecture: a physical quantum channel for key generation, a key-management plane with Key Management Entities (KMEs), and a conventional encrypted data plane. The ETSI GS QKD 014 REST API provides application-facing key delivery, while IKEv2 extensions for mixing preshared key material and performing multiple key exchanges offer protocol hooks for incorporating independent secrets [1]–[3]. The proposed hybrid key schedule combines classical, PQC-derived, and QKD-sourced contributions; PQC also supplies scalable authentication and works on ordinary IP paths.

The approach is a protocol-stack and architecture analysis connecting the design to CN1 topics including encapsulation, state machines, latency, throughput, error control, queuing, topology, and availability. Limited secret-key rate and KME key-pool exhaustion make buffering, rekey planning, admission control, and a defined PQC-only fallback important. Published key-delivery specifications, IKEv2 standards, and reported network testbeds support classifying QKD as an emerging technology with active standardization and limited niche deployment [1]–[4].

The analysis finds that key delivery and safe consumption in real protocol stacks—not key generation alone—are decisive integration issues. QKD can provide valuable additional protection on selected high-value links, but its reach, cost, device-security concerns, authentication bootstrap, and trusted-relay exposure limit deployment; relay nodes may hold key material in plaintext. Accordingly, the practical near-term design is hybrid: deploy PQC broadly for reach and authentication, and use QKD selectively where infrastructure and operational conditions justify it. The conclusion is defense in depth with explicit fallback and trust boundaries, not a universal QKD replacement for today's Internet security.

Keywords: quantum key distribution, post-quantum cryptography, IKEv2, key management, network security

REFERENCES

[1] "Quantum Key Distribution (QKD); Protocol and Data Format of REST-Based Key Delivery API," ETSI GS QKD 014 V1.1.1, European Telecommunications Standards Institute, Feb. 2019.

[2] C. Tjhai et al., "Mixing preshared keys in the Internet Key Exchange Protocol Version 2 (IKEv2) for post-quantum security," IETF RFC 8784, Jun. 2020.

[3] C. Tjhai et al., "Multiple key exchanges in the Internet Key Exchange Protocol Version 2 (IKEv2)," IETF RFC 9370, May 2023.

[4] "Module-Lattice-Based Key-Encapsulation Mechanism Standard," NIST FIPS PUB 203, National Institute of Standards and Technology, Aug. 2024.